Additional Statistical Evaluation of the Experimental Results

Table 1 shows the statistical evaluation using common metrics such as the average number of iterations until the optimal is found, standard deviation, median as well as lower and upper quartiles for the comparison between the TGP and CGP algorithms for various settings of D.

D	Model	Successful Runs	Avg. Iterations	SD	Min	Q1	Median	Q3	Max
1	simple_cgp_prob	30	4	6	0	0	0	5	27
	simple_cgp_sam	30	2	3	0	0	0	2	14
	simple_tgp_multi	30	4	4	0	1	2	7	15
	simple_tgp_single	30	4	4	0	0	3	7	13
2	simple_cgp_prob	30	27	26	1	14	24	31	146
	simple_cgp_sam	30	5	6	0	0	3	9	24
	simple_tgp_multi	30	33	23	3	16	26	49	86
	simple_tgp_single	30	20	12	4	11	19	26	49
3	simple_cgp_prob	30	95	78	2	48	76	107	349
	simple_cgp_sam	30	30	30	0	12	22	34	131
	simple_tgp_multi	30	117	91	12	69	96	125	489
	simple_tgp_single	30	85	33	25	65	82	110	160
4	simple_cgp_prob	30	244	177	31	120	174	312	789
	simple_cgp_sam	30	71	54	14	44	55	72	266
	simple_tgp_multi	30	257	79	150	197	251	319	415
	simple_tgp_single	30	199	60	108	160	197	228	347
5	simple_cgp_prob	30	560	442	42	245	438	794	2025
	simple_cgp_sam	30	131	73	53	70	109	160	342
	simple_tgp_multi	30	720	217	375	568	686	815	1351
	simple_tgp_single	30	557	239	241	395	483	676	1486
6	simple_cgp_prob	30	806	1014	213	346	530	796	5709
	simple_cgp_sam	30	249	172	57	110	204	320	711
	simple_tgp_multi	30	2036	839	1023	1571	1922	2164	5168
	simple_tgp_single	30	1345	379	844	1127	1290	1505	2363
7	simple_cgp_prob	30	1310	1264	349	522	882	1288	6000
	simple_cgp_sam	30	871	673	103	280	600	1334	2323
	simple_tgp_multi	30	4543	1012	2636	3802	4467	4856	6598
	simple_tgp_single	30	3282	834	2178	2718	3070	3634	5462
8	simple_cgp_prob	30	1877	1372	349	856	1578	2418	5916
	simple_cgp_sam	30	1786	1820	162	786	1336	2166	9535
	simple_tgp_multi	30	10791	3870	6607	8689	9406	12338	26357
	simple_tgp_single	30	7183	1731	4614	5997	6904	8172	12196
9	simple_cgp_prob	30	3982	3202	669	1774	2761	4986	13612
	simple_cgp_sam	30	2248	1624	218	1070	1764	3153	7101
	simple_tgp_multi	30	23624	4723	16106	20375	22840	25793	34053
	simple_tgp_single	30	17820	3124	13359	15315	16578	20385	23713
10	simple_cgp_prob	30	6122	4081	1971	3536	5406	7168	18905
	simple_cgp_sam	30	3687	3091	318	1289	3202	4687	13396
	simple_tgp_multi	30	53107	10589	34994	45680	53074	57450	75128
	simple_tgp_single	30	34477	5325	27017	31026	32674	38244	46205

Table 1: Results for the comparison between TGP and CGP on MAX for D up to 10.

Table 2 shows the statistical evaluation CGP of the average number of iterations until the optimal is found for experiments with large D up to 30. All runs are successful.

D	Model	Avg. Iterations	SD	Min	Q1	Median	Q3	Max
1	simple_cgp_prob	0	2	0	0	0	0	17
	simple_cgp_sam	8	11	0	0	4	10	67
2	simple_cgp_prob	18	19	0	5	13	25	99
	simple_cgp_sam	32	29	0	11	24	42	147
3	simple_cgp_prob	67	53	0	29	55	86	341
	simple_cgp_sam	81	58	12	43	64	104	375
4	simple_cgp_prob	200	156	41	94	159	260	792
	simple_cgp_sam	160	101	19	96	140	187	598
5	simple_cgp_prob	486	413	17	200	382	616	2029
	simple_cgp_sam	265	193	35	154	202	331	1319
6	simple_cgp_prob	724	556	73	345	544	985	3376
	simple_cgp_sam	399	255	72	211	328	517	1575
7	simple_cgp_prob	1223	874	204	606	930	1486	3712
	simple_cgp_sam	631	416	142	352	445	792	2283
8	simple_cgp_prob	1970	1520	273	744	1640	2661	7773
	simple_cgp_sam	916	644	222	509	678	1170	3764
9	simple_cgp_prob	2822	1613	543	1621	2373	4084	7794
	simple_cgp_sam	1504	1000	208	798	1270	1928	5036
10	simple_cgp_prob	4754	4757	653	1983	3726	5796	35394
	simple_cgp_sam	2138	1542	344	953	1744	3123	8651
11	simple_cgp_prob	6820	6719	899	2732	4638	7734	37599
	simple_cgp_sam	2402	2195	494	1242	1830	2948	18629
12	simple_cgp_prob	9023	9577	1296	3161	5774	10780	62620
	simple_cgp_sam	3307	2322	844	1728	2426	4292	14393
13	simple_cgp_prob	11105	8503	1388	5564	8446	14038	56769
	simple_cgp_sam	4719	3450	809	2480	3641	6212	21406
14	simple_cgp_prob	17824	14373	2767	8276	13956	22904	85134
	simple_cgp_sam	5565	3244	1661	3370	4638	7084	19020
15	simple_cgp_prob	18556	11493	2482	10069	15552	24349	59714
	simple_cgp_sam	6468	3609	1984	3748	5603	8259	18953
16	simple_cgp_prob	25033	23337	4842	10988	20026	31852	180966
	simple_cgp_sam	8450	5790	1788	4697	7204	9891	30595
17	simple_cgp_prob	28130	16214	5582	16337	22668	34384	84274
	simple_cgp_sam	9473	5252	1698	5798	8185	12665	25090
18	simple_cgp_prob	31842	17972	5230	18917	29416	39923	104998
	simple_cgp_sam	12812	7874	2197	7187	11518	15668	41348
19	simple_cgp_prob	41244	31837	10389	20629	29902	51697	187897
	simple_cgp_sam	14517	7653	4370	8394	13460	18424	47206
20	simple_cgp_prob	42229	26326	8849	24297	35874	51000	128573
	simple_cgp_sam	16404	9656	3777	10073	13880	20346	60878
21	simple_cgp_prob	54456	33793	7901	30814	47321	68663	184866
	simple_cgp_sam	20733	11303	3817	13083	19204	25310	73840
22	simple_cgp_prob	69422	47144	12259	37824	57672	83095	273118
	simple_cgp_sam	23365	13006	5971	13562	21096	29230	66820
23	simple_cgp_prob	70916	43429	16103	41690	60503	83953	259038
	simple_cgp_sam	28601	17026	5435	16684	23602	34954	79108
24	simple_cgp_prob	96044	64579	20201	55967	78390	116402	395964
	simple_cgp_sam	29319	16487	5261	18780	24965	36038	89190
25	simple_cgp_prob	89695	48531	27325	56280	79334	107248	298611
	simple_cgp_sam	35953	21309	6055	23612	31780	45032	167844
26	simple_cgp_prob	111358	59577	32305	69456	92536	145677	356496
	simple_cgp_sam	40979	19839	9143	27566	36811	49430	114584
27	simple_cgp_prob	133982	77459	24248	75764	117388	170679	436579
	simple_cgp_sam	42877	21408	12444	28741	38554	52671	123554
28	simple_cgp_prob	136296	99893	20717	64030	106946	156311	531601
	simple_cgp_sam	51730	36407	11516	33230	45726	60797	322109
29	simple_cgp_prob	170225	106375	31845	97621	151831	205370	672909
	simple_cgp_sam	52460	26374	12081	35366	45996	64015	154721
30	simple_cgp_prob	187986	109674	44497	104846	155190	246624	575885
	simple_cgp_sam	58458	27357	8476	37215	55200	74776	135409

Table 2: Results for the CGP comparison between `MutationProb` and `MutationSAM` for large D up to 30 using log-scaling.